need a notion of derivative

first variation of $\mathscr{F}$ at ρ

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- JKO Scheme Wasserstein space: $\rho^{k+1} \in \arg\min_\rho \ \mathscr{F}(\rho) \ + \ \frac{1}{2\tau} \mathrm{W}_2^2(\rho, \rho^k)$
- What happens as $\tau \to 0$?
- We **need a notion of derivative** of $\mathscr{F}(\rho)$ to study this
- For a functional $\mathscr{F} : \mathscr{P}(\mathbb{R}^d) \to \mathbb{R}$, the **first variation of $\mathscr{F}$ at ρ**, denoted $\frac{\delta \mathscr{F}}{\delta \rho}$ satisfies:

$$\frac{d}{d\varepsilon} \mathscr{F}(\rho + \varepsilon\chi)\Big|_{\varepsilon=0} = \int \frac{\delta \mathscr{F}}{\delta \rho}(x)\, \mathrm{d}\chi \qquad \forall \chi \in \mathscr{P}(\mathscr{X})\,.$$

- Intuition: $\frac{\delta \mathscr{F}}{\delta \rho}(x) \approx$ how much $\mathscr{F}$ changes if we add an infinitesimal amount of mass at location x

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